CLASS 10 - MATHEMATICS

Sample Student Answer Sheet

Student: Rahul Sharma Roll No.: 101

Q1

36 = 2 x 2 x 3 x 3

48 = 2 x 2 x 2 x 2 x 3

HCF = 2 x 2 x 3 = 12

Q2

3x - 7 = 11

3x = 18

x = 6

Q3(a)

Distance = sqrt((5-2)^2 + (7-3)^2)

= sqrt(9 + 16) = sqrt(25) = 5 units

Q4

Prime numbers on a die are 2, 3, 5.

Probability = 3/6 = 1/2

Q5

Circumference = 2 pi r
= 2 x 22/7 x 7 = 44 cm

Q6

Let numbers be x and y.
x + y = 27
x - y = 5
Adding: 2x = 32, so x = 16
y = 11

Q7

Using Pythagoras theorem:
base^2 + 8^2 = 10^2
base^2 = 36
base = 6 m

Q8

Mean = (6+8+10+12+14+16+18)/7
= 84/7 = 12

Q9

Curved surface area = 2 pi r h
= 2 x 22/7 x 3.5 x 10
= 220 cm^2

Q10(a)

x^2 - 7x + 12 = 0
(x - 3)(x - 4) = 0
x = 3 or 4

Q10(b)

a = 5, d = 3
20th term = a + 19d
= 5 + 57 = 62

Q11

Outer area = 30 x 20 = 600 m^2
Inner rectangle = 26 x 16 = 416 m^2
Area of path = 184 m^2
Cost = 184 x 120 = Rs. 22,080